

1. Introduction

In this class we explore Coulumb's Law using a few simple examples where we calculate the electric field generated by a complex charged body at any arbitrary position.

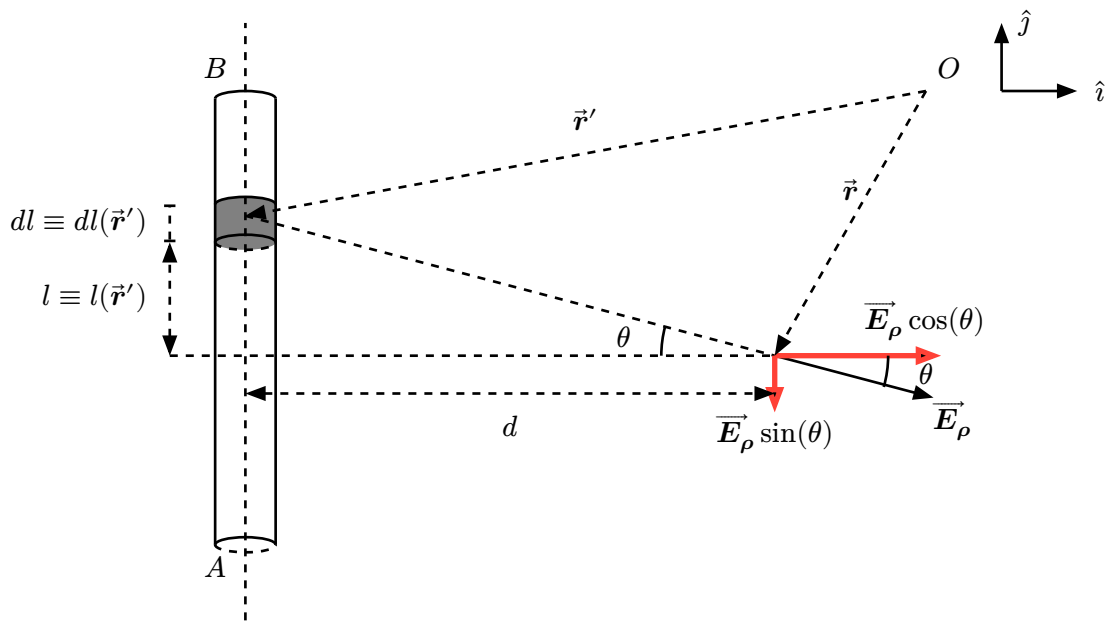


Figure 1: The figure illustrates a rod of finite length $\overline{AB}$ and negligible thickness with a charge density function (per unit length) ρ . An infinitesimally thin slice of the rod at position $\vec{r}'$ w.r.t the origin O is highlighted in grey. This highlighted slice of the rod is of thickness dl and l units away from the center of the rod. The components of the electric field along directions $\hat{i}$ and $\hat{j}$ at a position $\vec{r}$ which is d units away from the center of the rod has been highlighted in red.

Our goal in this class is to get familiar with Coulumb's Law and use it to calculate the electric field generated by some fairly complex charge distributions. Consider Figure 1, where we have a charged rod of length, say L , and a charge density function, say ρ . Our goal would be to calculate the electric field generated by this charged rod at the point represented by the position $\vec{r}$ w.r.t the origin O . As the rod is a fairly complex body with an arbitrary volume, we cannot consider it as a point charge. We need to deconstruct the rod into infinitesimally small parts and then use Coulumb's Law to calculate the the electric field generated by that small part. We can then aggregate the electric field contributions of all the parts and obtain the total electric field.

In the situation illustrated by Figure 1 we can consider an infinitesimally small cylinder of height $dl(\vec{r}')$ located at position $\vec{r}'$ w.r.t the origin O . Now, let us denote the infinitesimally small electric field generated by this cylinder at position $\vec{r}$ by $d\vec{E}_{\vec{r}'}(\vec{r})$. Using Coulumb's Law we can write

$$d\vec{E}_{\vec{r}'}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho dl(\vec{r}')$$

Now to calculate the electric field generated by the complete rod we integrate over $\vec{r}'$ from $\vec{r}_A$ to $\vec{r}_B$. This can be written as

$$\begin{aligned}\vec{E}_\rho(\vec{r}) &= \int_{\vec{r}_A}^{\vec{r}_B} d\vec{E}_{\vec{r}'}(\vec{r}) \\ &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho dl(\vec{r}')$$

The calculation is fairly simple and is provided below

$$\begin{aligned}\vec{E}_\rho(\vec{r}) &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} dl(\vec{r}') \\ &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\cos(\theta) dl(\vec{r}')}{|\vec{r} - \vec{r}'|^2} \hat{i} + \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\sin(\theta) dl(\vec{r}')}{|\vec{r} - \vec{r}'|^2} \hat{j} \\ &= \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{4\pi\epsilon_0} \rho \frac{\cos(\theta) dl}{l^2 + d^2} \hat{i} + \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{1}{4\pi\epsilon_0} \rho \frac{\sin(\theta) dl}{l^2 + d^2} \hat{j} \\ &= \hat{i} \frac{\rho d}{4\pi\epsilon_0} \int_{-\frac{L}{2}}^{\frac{L}{2}} \frac{dl}{(l^2 + d^2)^{\frac{3}{2}}} \hat{i} \\ &= \hat{i} \frac{\rho d}{4\pi\epsilon_0} \left[\frac{l}{d^2 \sqrt{l^2 + d^2}} \right]_{l=-\frac{L}{2}}^{l=\frac{L}{2}} \\ &= \hat{i} \frac{\rho d}{4\pi\epsilon_0} \left[\frac{\frac{L}{2}}{d^2 \sqrt{(\frac{L}{2})^2 + d^2}} + \frac{-\frac{L}{2}}{d^2 \sqrt{(-\frac{L}{2})^2 + d^2}} \right] \\ &= \hat{i} \frac{\rho}{4\pi\epsilon_0 d} \left[\frac{L}{\sqrt{(\frac{L}{2})^2 + d^2}} \right]\end{aligned}$$

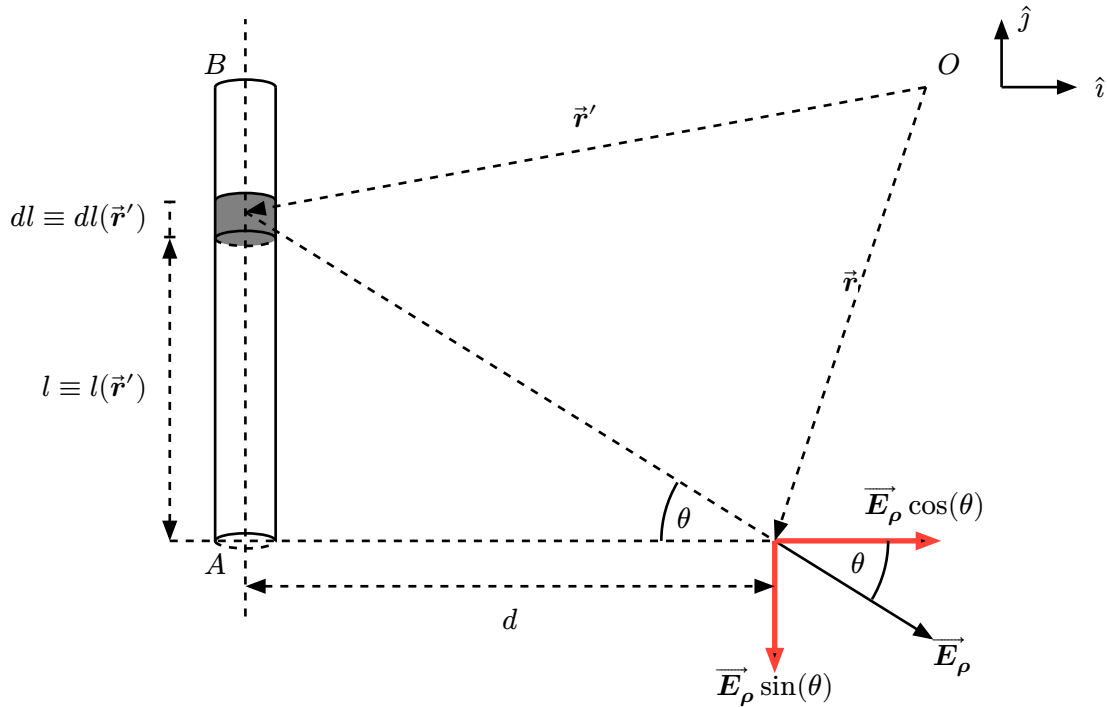


Figure 2: The figure illustrates a rod of finite length $\overline{AB}$ and negligible thickness with a charge density function (per unit length) ρ . An infinitesimally thin slice of the rod at position $\vec{r}'$ w.r.t the origin O is highlighted in grey. This highlighted slice of the rod is of thickness dl and l units away from one end of the rod. The components of the electric field along directions $\hat{i}$ and $\hat{j}$ at a position $\vec{r}$ which is d units away from one end of the rod has been highlighted in red.

Let us consider a second example, Figure 2, where we have a charged rod of length, say L , and a charge density function, say ρ . Our goal would be to calculate the electric field generated by this charged rod at the point represented by the position $\vec{r}$ w.r.t the origin O . Similar to the previous case, the rod is a fairly complex body with an arbitrary volume, we cannot consider it as a point charge. We need to deconstruct the rod into infinitesimally small parts and then use Coulomb's Law to calculate the electric field generated by that small part. We can then aggregate the electric field contributions of all the parts and obtain the total electric field.

In this new situation illustrated by Figure 2 we can consider an infinitesimally small cylinder of height $dl(\vec{r}')$ located at position $\vec{r}'$ w.r.t the origin O . Now, let us denote the infinitesimally small electric field generated by this cylinder at position $\vec{r}$ by $d\vec{E}_{\vec{r}'}(\vec{r})$. Using Coulomb's Law we can write

$$d\vec{E}_{\vec{r}'}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho dl(\vec{r}')$$

Now to calculate the electric field generated by the complete rod we integrate over $\vec{r}'$ from $\vec{r}_A$ to $\vec{r}_B$. This can be written as

$$\begin{aligned} \vec{E}_\rho(\vec{r}) &= \int_{\vec{r}_A}^{\vec{r}_B} d\vec{E}_{\vec{r}'}(\vec{r}) \\ &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} \rho dl(\vec{r}') \end{aligned}$$

The calculation is fairly simple and is provided below

$$\begin{aligned}
 \vec{E}_\rho(\vec{r}) &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\vec{r} - \vec{r}'}{|\vec{r} - \vec{r}'|^3} dl(\vec{r}') \\
 &= \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\cos(\theta) dl(\vec{r}')}{|\vec{r} - \vec{r}'|^2} \hat{i} + \int_{\vec{r}_A}^{\vec{r}_B} \frac{1}{4\pi\epsilon_0} \rho \frac{\sin(\theta) dl(\vec{r}')}{|\vec{r} - \vec{r}'|^2} \hat{j} \\
 &= \int_0^L \frac{1}{4\pi\epsilon_0} \rho \frac{\cos(\theta) dl}{l^2 + d^2} \hat{i} + \int_0^L \frac{1}{4\pi\epsilon_0} \rho \frac{\sin(\theta) dl}{l^2 + d^2} \hat{j} \\
 &= \frac{\rho d}{4\pi\epsilon_0} \int_0^L \frac{dl}{(l^2 + d^2)^{\frac{3}{2}}} \hat{i} + \frac{\rho}{4\pi\epsilon_0} \int_0^L \frac{l dl}{(l^2 + d^2)^{\frac{3}{2}}} \hat{j} \\
 &= \hat{i} \frac{\rho d}{4\pi\epsilon_0} \left[\frac{l}{d^2 \sqrt{l^2 + d^2}} \right]_{l=0}^{l=L} + \hat{j} \frac{\rho}{4\pi\epsilon_0} \left[-\frac{1}{\sqrt{l^2 + d^2}} \right]_{l=0}^{l=L} \\
 &= \hat{i} \frac{\rho}{4\pi\epsilon_0 d} \left[\frac{L}{\sqrt{L^2 + d^2}} \right] + \hat{j} \frac{\rho}{4\pi\epsilon_0} \left[\frac{1}{d} - \frac{1}{\sqrt{L^2 + d^2}} \right]
 \end{aligned}$$